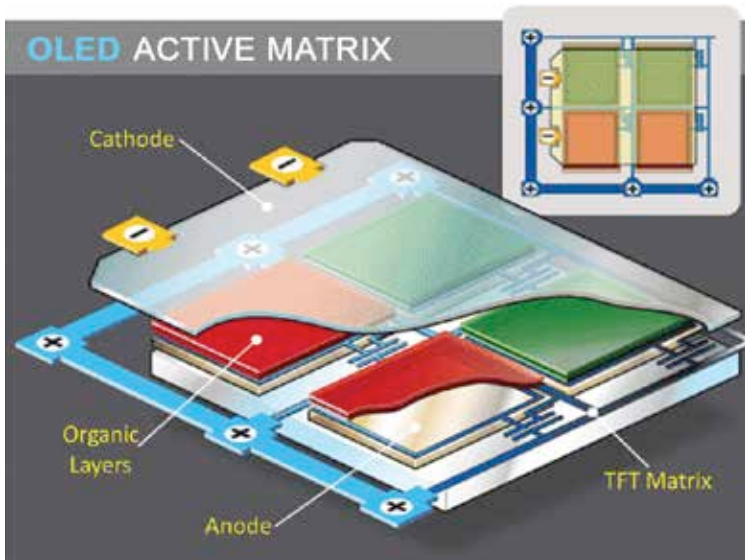


The new generation: Organic LEDs

The use of OLEDs – organic LEDs – is ushering in a different era of largely scalable displays that consume almost half the power conventional LED screens and LCDs do. While an OLED is similar to an LED, it uses a layer of organic molecules that have the ability to become illuminated when the molecules are ionised. Unlike LCD (Liquid Crystal Display) technology, OLED does not require backlighting to function. The internal structure of an OLED screen is basically a layer of this organic material sandwiched



Cross section of a typic OLED panel

between an anode and cathode, with another layer of conducting material to boot. The anode, which is positively charged, pulls out electrons from the conducting layer, leaving the organic layer to be sandwiched between two layers of positively (the conducting layer) and negatively charged (cathode) particles. Hence, this eliminates the need for any backlighting, reducing the weight and thickness of the screen by half, and further cutting down on power expenditure.

The most commonly used OLED screen is the AMOLED – an active matrix OLED. Except for the technique of powering the LED's (sub-pixel). AMOLED cells contain the above described structure of an OLED screen